

What is the Internet doing to me?
(witidtm 2026/2027 - TEU00311)

Lab Sessions

Stephen Farrell
stephen.farrell@cs.tcd.ie

<https://github.com/sftcd/witidtm>
<https://down.dsg.cs.tcd.ie/witidtm>

People

- We've some helpers today, feel free to grab any of us and ask stuff
- As well as me and Abeba, there's:
 - Matias Duran
 - Riccardo Angius

Lab#1

Things for today...

- Login to SCSS account on lab machine and/or get your laptop online
- GOTO: <https://duck.ai/chat> and play about a bit
- Do Abeba's assignment whilst here...

Login/get-online

- Desktop logins: no “domain”, use SCSS password, not your TCD password (if those differ), so e.g. if your TCD email is **bloggsj14@tcd.ie** then you enter “bloggsj14” as the username
- Get online: we’ll deal with things as they arise
- After you’re done: see if anyone else needs help

I did check that username... :-)

“Your message to bloggsj14@tcd.ie couldn't be delivered.

A custom mail flow rule created by an admin at tcdud.onmicrosoft.com has blocked your message.

5.1.1 The e-mail service at tcd.ie does not know this email address.”

Large Language Models (LLMs)

- ChatGPT and pals are quite popular but interestingly unreliable, so let's play with that
- DuckDuckGo provide an interface to interact with various LLMs that doesn't require an account:
 - <https://duckduckgo.com/aichat>
- We do not know if all of us using that at once might hit some rate limiting (hopefully not, but we'll see...)

Differences in size / energy consumption

- Models show with a "B" how many billion parameters they have.
- Generally, **+parameters, more capable, higher cost**
- Easy tasks do not need the biggest models and using them would be a waste, bigger models generally consume more energy.
- This is what ChatGPT-5 changed, making the decision for each prompt on what style of model to use.
- **Quantization** reduces energy usage while maintaining number of parameters by reducing resolution of each parameter.

Ref: <https://ieeexplore.ieee.org/document/10549890/>
<https://arxiv.org/pdf/2507.11417>

Models learn a distribution, reproduce bias

- "African people at work" vs "European people at work"
- "Emotional author" vs "intellectual author" (try yourselves)



Asking for African first and then modifying to European.



Asking for European first and then modifying to African .

Go to: <https://duckduckgo.com/aichat>

Pick a model

Read the T&C's

Play...

Some prompts to try/modify

- How many r's in raspberry?
 - You may need to try different models to get the wrong answer:-)
- **...more TBD...**

AI as a search tool

“AI Search Has A Citation Problem”, March 2025, Columbia Journalism Review, Klaudia Jaźwińska and Aisvarya Chandrasekar

- https://www.cjr.org/tow_center/we-compared-eight-ai-search-engines-theyre-all-bad-at-citing-news.php
- “We Compared Eight AI Search Engines. They’re All Bad at Citing News”
- Confidently incorrect/speculative answers; fabricated links, etc.

Lab#2

(Slides likely to change
from here on...)

Browsers and the Web

- Sort yourselves into groups of ~2-3
- Fire up a “new”/“clean” browser
- Explore settings, esp security/privacy related
- Watch HTTP traffic (shift-ctrl-I)
- Find the “worst” site you can
 - NSFW disallowed!!!
 - You define/justify “worst”
 - Report back from group

Stretch goal

- If we get the above done great, if not, that's ok
- Could be some people get to the stretch goal today or we look at it in future...
- Stretch goal: HTTP archive (.har) file generation and a bit of analysis

Play with “new” browser

- Don't use one that has e.g. stored credentials for some account you care about – basically don't muck up your daily-driver setup
- Lab machine browsers can probably be reset easily enough (TBC)
- On own device: maybe install one you've not used before
 - Possibles: Brave, Firefox, chromium, vivaldi, opera...
 - Tor Browser and/or Brave supports opening a window via Tor
 - More exist, but start being careful if you go for something less widely used: browser downloads have been known to contain malware from time to time (but mostly on phones)
- On mobile devices you might also try the DuckDuckGo Browser:
 - <https://duckduckgo.com/duckduckgo-help-pages/mobile/ios>

Browser settings

- Defaults are important and not always what *you* would want!
- Play about in settings and see what you find – ask if not sure or comment if you think you should tweak
- As they differ a bit, and change over time, you might want do this now and then for all browsers you use: e.g. lab machine, laptop, phone
- Things to check/set:
 - Telemetry/sending usage info
 - Search engine
 - Search/URL bar suggestions
 - AI controls
 - Locations/Camera/
Microphone/Notification
permissions
 - App launching
 - Private DNS/DNS over
HTTPS (DoH)
 - Home/new page settings
 - Cookies
 - Blockers/Tracking protection
 - Logins/passwords/Payment
methods
 - Javascript
 - “HTTPS-only” mode
 - Clear things on exit
 - Stored site permissions

Watch web traffic

- Open browser
- Type shift-ctrl-I (or equivalent) to open developer interface
- Re-size screens to taste
- Choose “network” tab in developer pane
- Try loading a few sites and watch what happens
 - DO NOT load NSFW sites!
- Look about on the web and decide which site is the “worst” from your POV and why
 - Just yell/put up hand when you have a “worst” to nominate
 - We’ll pick a winner if we’ve time – Prize == applause:-)

Shift-ctrl-I for macs...

- Macs differ:
 - Firefox: Option + Command + I
 - Safari: Option + Command + C
 - Chrome: Option + Command + C
- Access to developer tools in Safari has to be activated in the settings first. If anyone has problems with that they can find detailed information on how to do this here: <https://support.apple.com/en-ie/guide/safari/sfri20948/mac>
- Thanks to Luca Schäfer (2021 student) for the above

Browser Extensions

- Most browsers have a library of add-ons or extensions that you can download and install
- Some are quite useful:
 - Firefox: NoScript – compare page load times with/without
 - Others?
- Note: be a bit careful – extensions are powerful things and can see your browsing!

Stretch goal

- Figure out how to save an HTTP archive file (.har)
- Figure out how to view .har files
- Figure out how to diff .har files
- See what changes between seemingly identical browser sessions
 - ...any tracking?

Go do stuff!

Images

- Find the location of an image
- Consider what facial recognition means for us
- Stretch goal: minimal image manipulation to defeat recognition
- DO NOT use any image that has a reasonable probability of upsetting anyone

Images and the Internet

- Why are details of images relevant to this module?

Why are details of images relevant to this module?

- We upload lots of images
- Automated image capture is near ubiquitous
 - CCTV, ANPR, ...
- Other people upload images of us
- Organisations with image databases analyse those

Data vs. Metadata

- Typically we talk about the “data” as being the main thing being processed or communicated or stored...
 - E.g: the bits rendered for an image or video, the content of an email, the messages in a text chat or the audio packets in a voice call
- So-called “metadata” is also data but is “about” the above rather than part of the above
 - E.g. timing of a communication, sender/receiver IP addresses, the size of data fields, etc
- Even if data is well-protected (e.g. encrypted), metadata can leak separately (or be deliberately stored/exposed) so meta-data creates risk
- For someone surveilling, metadata can be more attractive than data, e.g. law enforcement may benefit more from building a social graph of criminals compared to seeing the content of a few messages, or, facebook might learn enough from whatsapp metadata that they no longer need to see the content to sell advertising
- Metadata is also often more structured and hence easier to process for those who want to see what people are doing
- Metadata can also be a little unexpected to end-users, e.g. author information in documents, or, in images...

Image Metadata: EXIF

<https://photographylife.com/what-is-exif-data>

<https://helpdeskgeek.com/how-to/how-to-remove-exif-data-from-your-photos/>

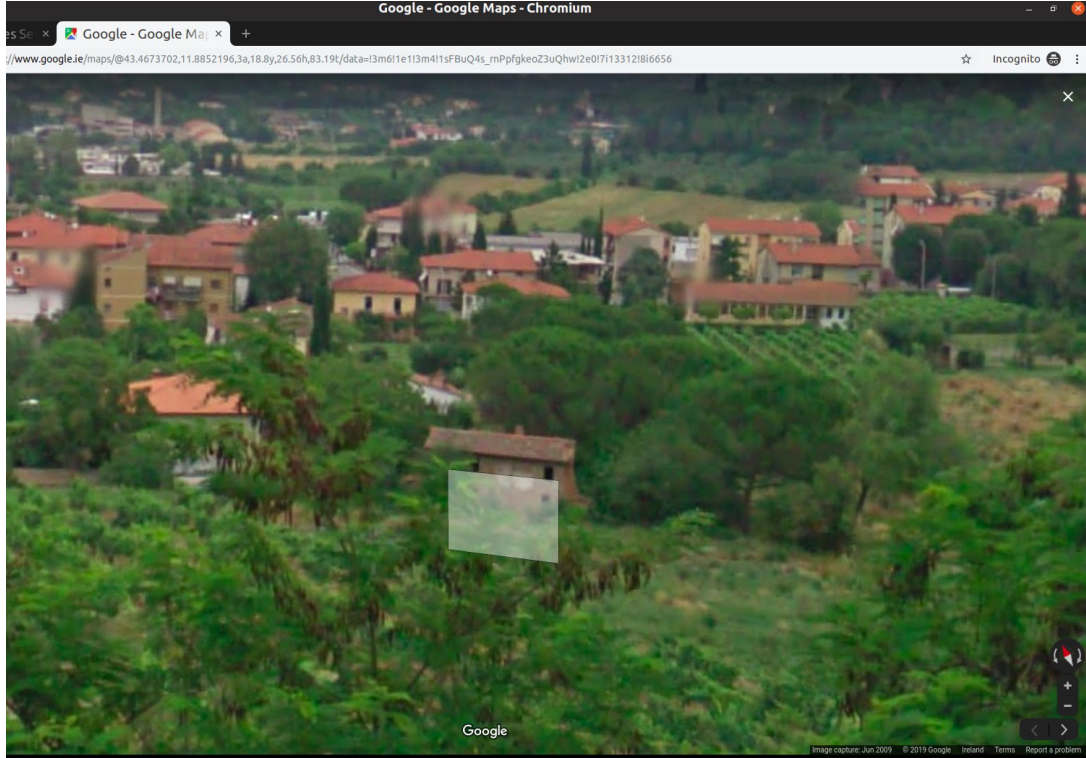


```
exif DSCN0010.jpg
EXIF tags in 'DSCN0010.jpg' ('Intel' byte order):
-----
Tag                |Value
-----
Image Description   |
Manufacturer       |NIKON
Model              |COOLPIX P6000
Orientation         |Top-left
X-Resolution        |300
Y-Resolution        |300
Resolution Unit     |Inch
Software            |Nikon Transfer 1.1 W
Date and Time       |2008:11:01 21:15:07
YCbCr Positioning   |Centered
Compression         |JPEG compression
X-Resolution        |72
Y-Resolution        |72
Resolution Unit     |Inch
Exposure Time       |1/75 sec.
F-Number            |f/5.9
Exposure Program    |Normal program
ISO Speed Ratings    |64
Exif Version        |Exif Version 2.2
Date and Time (Orig)|2008:10:22
16:28:39
Date and Time (Digit)|2008:10:22 16:28:39
Components Configura|Y Cb Cr -
Exposure Bias       |0.00 EV
Maximum Aperture Val|2.00 EV (f/2.7)
Metering Mode       |Pattern
Light Source         |Unknown
Flash               |Flash did not fire, compulsory flash mode
```

```
Focal Length        |24.0 mm
Maker Note          |3298 bytes undefined data
User Comment        |
FlashPixVersion     |FlashPix Version 1.0
Color Space         |sRGB
Pixel X Dimension    |640
Pixel Y Dimension    |480
File Source         |DSC
Scene Type          |Directly photographed
Custom Rendered     |Normal process
Exposure Mode       |Auto exposure
White Balance        |Auto white balance
Digital Zoom Ratio   |0.00
Focal Length in 35mm|112
Scene Capture Type   |Standard
Gain Control         |Normal
Contrast             |Normal
Saturation           |Normal
Sharpness           |Normal
Subject Distance Ran|Unknown
North or South Latit|N
Latitude            |43, 28,
2.81400000
East or West Longitu|E
Longitude           |11, 53,
6.45599999
Altitude Reference   |Sea level
GPS Time (Atomic Clo)|14:27:07.24
GPS Satellites       |06
GPS Image Direction  |
Geodetic Survey Data|WGS-84
GPS Date            |2008:10:23
Interoperability Ind|R98
Interoperability Ver|0100
-----
EXIF data contains a thumbnail (6702 bytes).
```

<https://raw.githubusercontent.com/ianare/exif-samples/master/jpg/gps/DSCN0010.jpg>

52100 Arezzo, Province of Arezzo, Italy



- Took about 5 minutes to find this in Google street view
- Most of that was finding a way to map degree, minutes, seconds to fractional Lat,Long
- All **automatable**, could easily produce location history from a set of images
- How could such a “leak” be damaging to you or to someone else in your images?

Viewing EXIF Data

- Local: Right-click and “properties”
- Better local: install something allowing you to scan multiple images
 - E.g. “`sudo apt install libimage-exiftool-perl`” on Ubuntu
- On web: save image to local then GOTO above

My Pictures

- A couple of years back, I scanned the 35,429 files in my Pictures directory and found ~493 of those (excl. dups) contained EXIF GPS lat/long values
 - Earliest was from 2009, most recent from 2022
 - Latitude ranged from 67° N, to 3° S
 - Longitude ranged from 157° W, to 67° E
 - The 157° W pic: a conference in Hawaii I attended remotely;-(
- On the linux command line that looked like:

```
$ exiftool -gpslatitude -gpslongitude -createdate -csv -r ~/Pictures
```

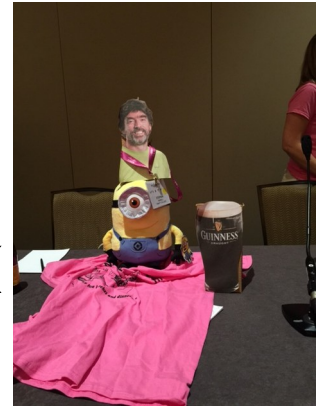


Photo Sharing (1)

- What happens when you share photos?
 - Worth noting that some people, esp. photo enthusiasts, do want all the EXIF data preserved/shared
- “Approximately 69% of services under study were found to actively sanitize EXIF metadata, which mitigates privacy risks for the users.”
- The quote above and the table on the next slide are from:
 - Gouert, Charles, and Nektarios Georgios Tsoutsos. "Dirty Metadata: Understanding A Threat to Online Privacy." IEEE Security & Privacy 20.6 (2022): 27-34.
<https://udspace.udel.edu/server/api/core/bitstreams/f2a04ac2-e998-49a4-b943-c26659c74fe4/content> accessed 20250916
- If “scrubbing” happens on the server, the server still gets metadata even if service users don’t, not clear when that might apply
- A recent (April 2026) blog on that (not sure how accurate):
 - <https://fast.io/resources/social-media-photo-metadata-platforms-strip/>

Photo Sharing (2)

Service	Class	Platforms	Policy	Service	Class	Platforms	Policy
Signal	Messaging	iOS, Windows	Scrub	420chan	Imageboard	Browser	Scrub
FB Messenger	Messaging	iOS, Browser	Scrub	2chan	Imageboard	Browser	No Scrub
iMessage	Messaging	iOS	No Scrub	Hispachan	Imageboard	Browser	Scrub
Slack	Messaging	Ubuntu, iOS	Scrub	Diochan	Imageboard	Browser	Scrub
Skype	Messaging	Ubuntu, Windows	Partial Scrub	Ylilauta	Imageboard	Browser	Scrub
WhatsApp	Messaging	iOS	Scrub	MS Outlook	Email	Windows	No Scrub
Viber (PC)	Messaging	Ubuntu	No Scrub	Yahoo Mail	Email	Browser	No Scrub
Viber (Phone)	Messaging	iOS	Scrub	Gmail	Email	Browser	No Scrub
WeChat	Messaging	iOS	Scrub	Protonmail	Email	Browser	No Scrub
Instagram	Social Media	Browser, iOS	Scrub	AOL Mail	Email	Browser	No Scrub
Twitter	Social Media	Browser	Scrub	Craigslist	Marketplace	Browser	Scrub
Facebook	Social Media	Browser	Scrub	Amazon	Marketplace	Browser	Scrub
LinkedIn	Social Media	Browser	Scrub	eBay	Marketplace	Browser	Scrub
Reddit	Imageboard	Browser	Scrub	FB Marketplace	Marketplace	Browser	Scrub
4chan	Imageboard	Browser	Scrub	Alibaba	Marketplace	Browser	No Scrub
PTchan	Imageboard	Browser	Scrub	Etsy	Marketplace	Browser	Scrub
Flickr	Imageboard	Browser	No Scrub	Tanga	Marketplace	Browser	Scrub

Your EXIF task...

- 1) Find some image(s) online or locally
 - 2) Determine if they contain EXIF location data
 - 3) Find the location of that image in e.g. Google street view
 - 4) As time permits: GOTO 1
-
- What can you infer from the above?
 - What could you infer if you did the above for a number of images of related subjects?

Go do stuff!

Facial recognition

- A kind of “biometric” (more later on the imperfections of biometrics:-)
- Nice overview, including tricky issues at: https://en.wikipedia.org/wiki/Facial_recognition_system (accessed 20250916)
- Basic idea: program analyses image bits, search for pattern that looks like a face (eyes, nose, mouth, ...), classifies that (based on machine learning using image collections), compare results from two images – if close enough, declare match
 - False positives and negatives will happen
- Note: this is not my area of expertise!

Facial recognition (ab)uses

- Find a photo of “this person”
 - Find local pics of your mum, organise your image gallery
- Find people with outstanding arrest warrants in a crowd
 - Recognition of faces in moving crowd is harder than individually, but likely, not that much harder
- Determine ethnicity of people using public transport
 - Critics may say things like the above - proponents might talk about improving efficiency but build systems that have this effect

Surveillance AI Pipeline

Paper by Abeba: <https://www.nature.com/articles/s41586-025-08972-6>

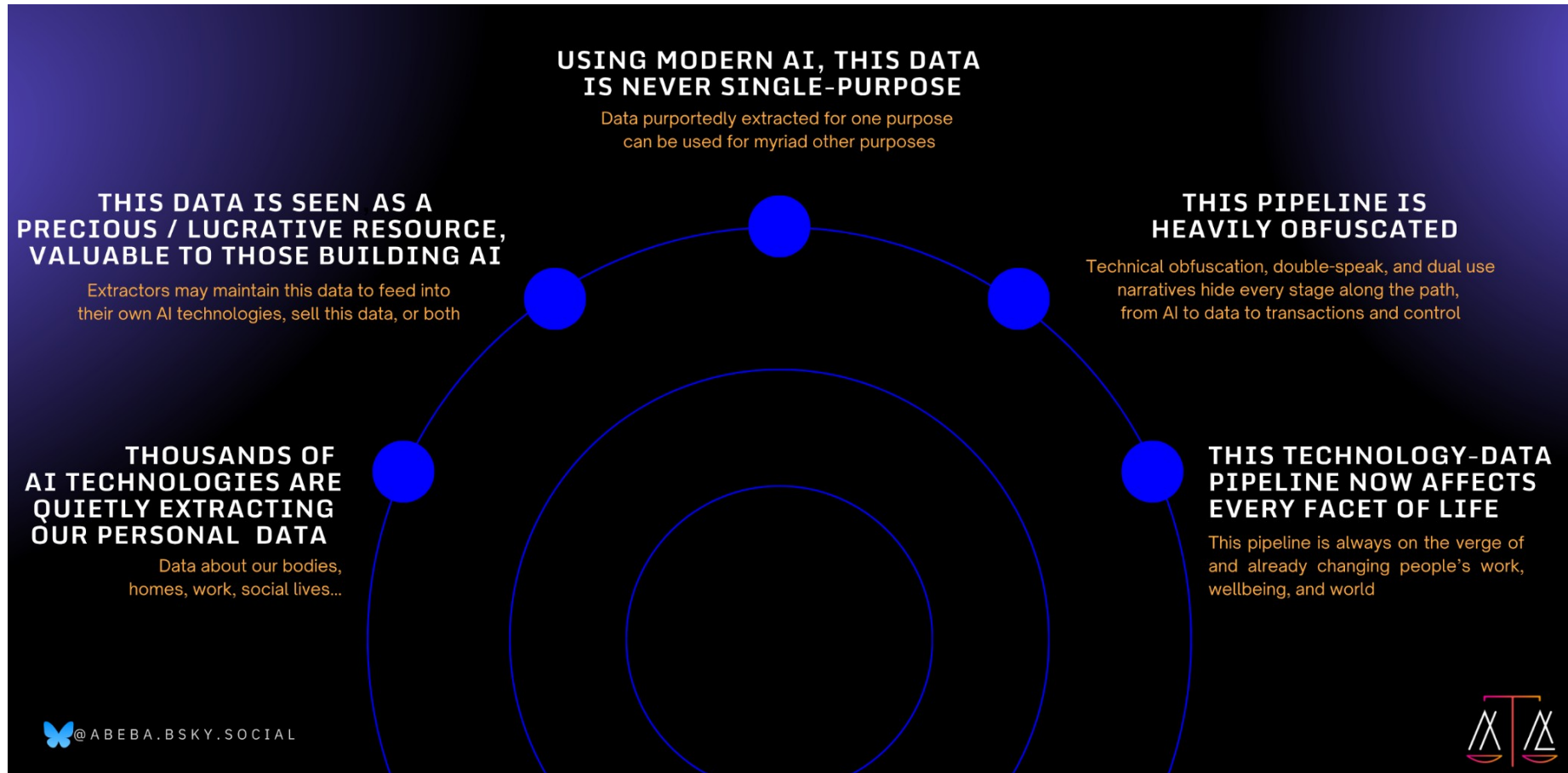


Image Search

- Google “reverse” image search
 - GOTO <https://images.google.com/>
 - Click the “camera” icon
 - Paste/upload an image
 - Google will search for info and other images

Image Search Task

- Your task: play with that image search to try find images that produce bad search results
 - The goal is to get a feel for what does/doesn't work well with that
- Don't use any NSFW images
- Don't use images of people who might not want you to use their image (including yourself if that's the case!)

Stretch goal – Try break that

- Find an image that produces good search results (easy:-)
- Pick an image editor
 - My suggestion: gimp, <https://www.gimp.org/>
 - But there may be simpler options
- Try find the “smallest” change that causes the search results to be noticeably worse
- General topic: adversarial images
 - <https://christophm.github.io/interpretable-ml-book/adversarial.html> accessed 20250916
 - Same concepts apply to other machine learning settings, e.g. text, audio, ...

Adversarial Example

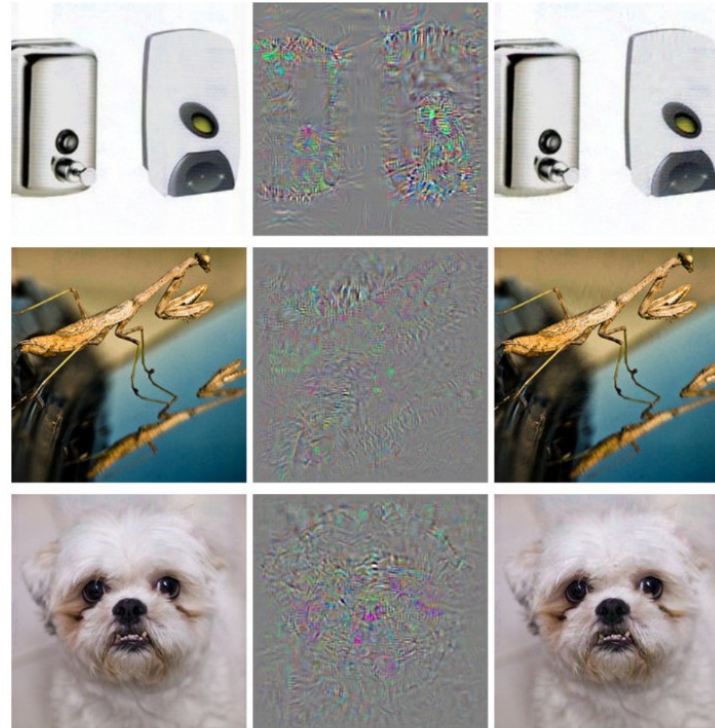


Image is shown at <https://christophm.github.io/interpretable-ml-book/adversarial.html> and originally from:

Szegedy, Christian, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. 2014. "Intriguing Properties of Neural Networks." arXiv. <https://doi.org/10.48550/arXiv.1312.6199>.

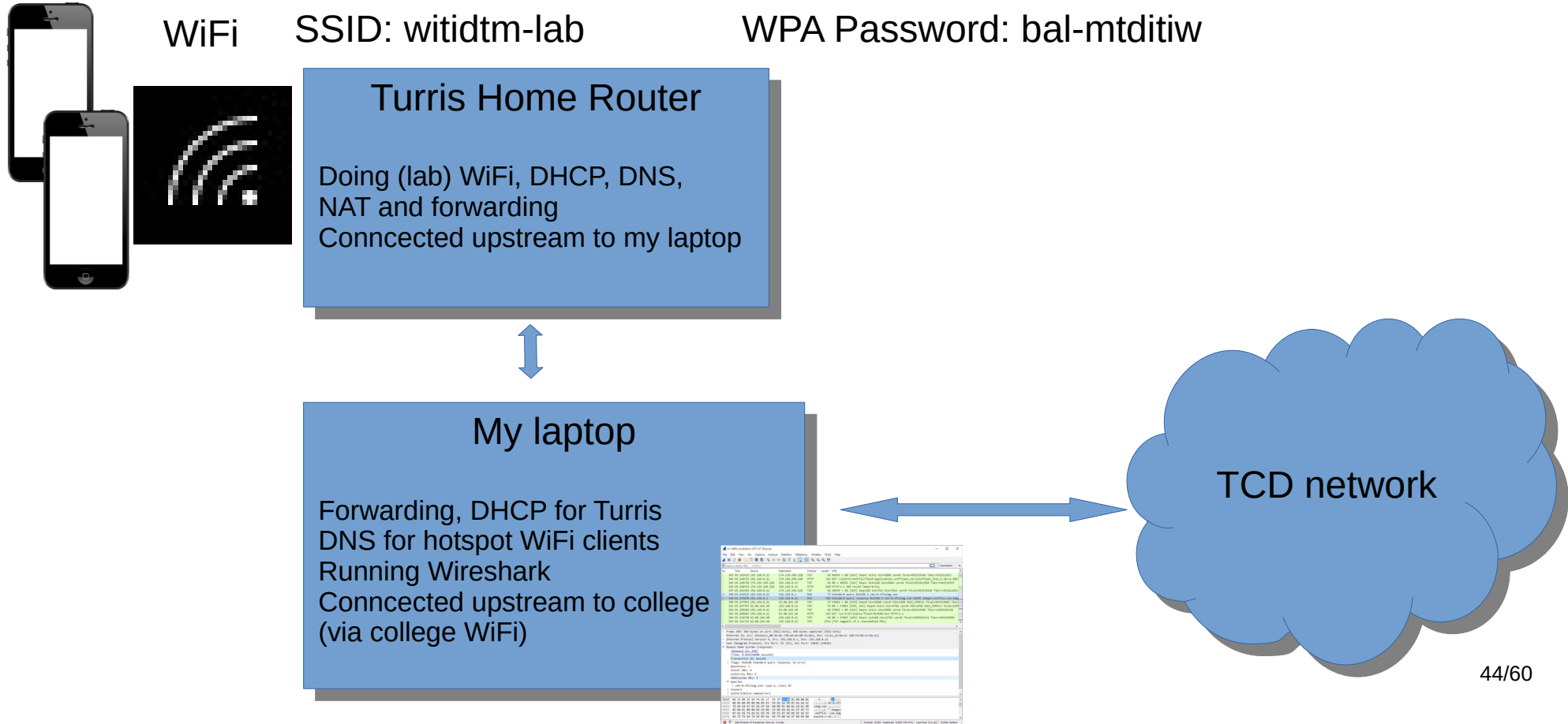
Figure 30.1: Adversarial examples for AlexNet by Szegedy et al. (2013). All images in the left column are correctly classified. The middle column shows the (magnified) error added to the images to produce the images in the right column all categorized (incorrectly) as "Ostrich". "Intriguing properties of neural networks", Figure 5 by Szegedy et al. CC-BY 3.0.

Go do stuff!

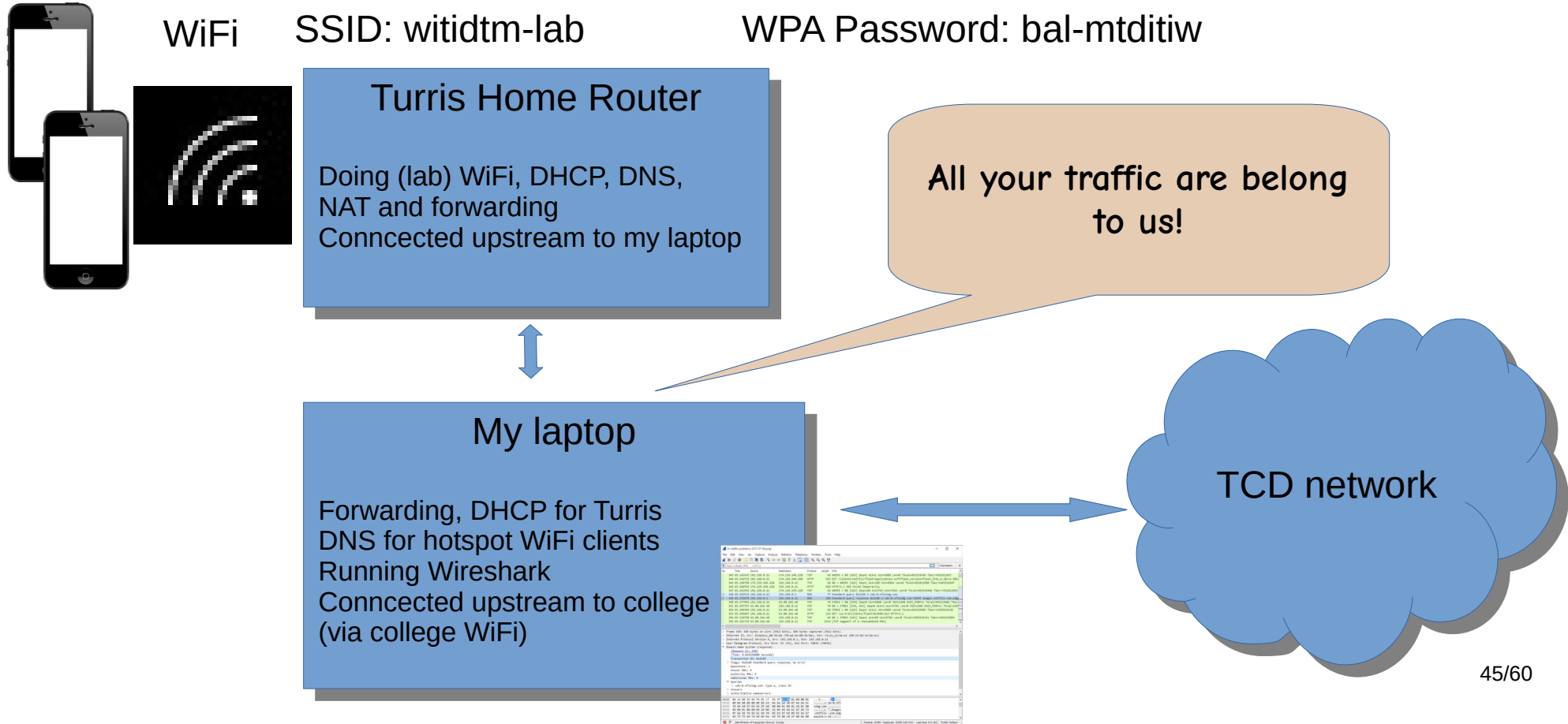
Network Obsevers

- Get brave/wireshark 1st!
- Join my hotspot
 - SSID: witidtm-lab
 - WPA Password: bal-mtditiw
- We'll watch some traffic using wireshark, and chat about that
- Stretch goal: do it yourself at home if interested

Hotspot setup



Hotspot setup



Wireshark

- Fine tool to observe network traffic
 - <https://www.wireshark.org/>
- The network-layer equivalent of what you saw in a browser with shift-ctrl-I
- But not limited to web traffic, you see it all as it is “on the wire”

Some traffic

- Display wireshark and see what's visible

MAC Addresses

- Device-tracking often (ab)uses **long-term hard-coded identifiers** such as MAC addresses (or IMEI/IMSI in mobile n/w)
- MAC address: link layer address (mostly) hardcoded to radio or other network chip
 - Same form of address used in WiFi and most other network protocols at link layer, e.g. Bluetooth
 - Roughly: how two devices on the same local area network (LAN) identify one another
- Looks like “6C:9C:ED:87:27:60” (48 bits) - 1st half is manufacturer ID (Cisco), 2nd half device-ID (a WiFi router that is/was in TCD SCSS)
 - You can look up manuf IDs from the registry, e.g. <https://www.macvendorlookup.com/> accessed 20250916
- MAC address is often fixed for the lifetime of the device; There is now a 64-bit version, not sure how widely used yet
 - You can probably see these in your “about device” tab or similar

Randomised MACs

- MAC address randomisation is a good idea and now fairly well deployed
 - Often, the MAC address only really needs to be stable for a session, so can be randomised
 - But – if you paid for the hotel WiFi that might be based on your MAC address, or an enterprise network might use MAC addresses to decide which machines are allowed on the local network, or the machine may be a switch/router/server where changing MAC address would break stuff or be inefficient
- So you can't always randomise, and doing so well needs higher-layer controls
- HOWTO turn on varies by OS and version
- On an android 10 phone I used have:
 - Developer options/Enhanced Wi-Fi MAC randomisation
 - You may need to turn on developer options first (search for HOWTO)

Some traffic

- Display wireshark and see what's visible

DHCP (1)

- Dynamic Host Configuration Protocol (DHCP) is (almost always) how your device gets an IPv4 address after joining a network
 - Spec is RFC2131 from 1997
- DHCP has a “hostname” option that client’s send and that has often been the same as a long-term device name, e.g. “Stephen’s iPhone”
 - That can be used for a loooong time
 - Recent phone OSes tend to send something more random looking but often don’t vary the value
 - If you migrate from an old to a new device, that setting might carry over even if your current OS would otherwise use a random value
 - Changing hostname could break stuff though so be a bit careful
- Other DHCP options clients send can also be identifying e.g. OS version

DHCP (2)

- As well as returning an IP address, the DHCP server can send many other options
 - Most aren't widely deployed
- The DNS server IP address option is though
 - Tells clients what server to use for DNS in this network
- OSes can override that but mostly (so far) don't
 - Unless you have DoT configured or some Apple stuff

Router Admin

- Demo OpenWRT/Foris/Luci

DNS names

- Today, we can mostly see the DNS names being queried as that's cleartext
- Starting to see more use of encrypted DNS traffic
- Two flavours: DoT and DoH
- We'll look quickly at that but chat more about it later
- On that old crappy android device enabling DoT used be:
 - Settings/Connections/More Connection Settings/Private DNS
 - BUT that won't work with college as upstream as DoT uses port 853 and college block that port – it should work fine at home and with your mobile data provider

Do53 vs DoT vs DoH

- Do53 == old style cleartext DNS
- Who do you want to/care about seeing your DNS traffic?
 - Your ISP, TCD, coffee-shop and their ISP, Cloudflare, Google ?
- Pros and cons to each of these

Brave Browser

- Not a bad browser on mobiles
- Has various “shields up/down” settings
 - Settings/Brave Shields and Privacy/Use Secure DNS
 - DoH – that does work in college
 - There may be a set of known services from which to choose
 - Or you can add a custom one:
 - For cloudflare try: <https://1.1.1.1/dns-query>
 - DoH can also be set in most other browsers these days too

Some traffic

- Display wireshark and see what's visible

Stretch Goal

- Repeat this at home if interested
 - Can help with, but not mandatory for, my assignment
- Setup hotspot using laptop
- Install wireshark
- Inspect traffic and learn

http:// vs. https://

- https:// URLs turn on encryption and web-site authentication between your browser and the web site
 - Far from a perfect system, but good enough 90%+ of the time
 - 98% of web traffic as seen by cloudflare uses https:// URLs
 - <https://radar.cloudflare.com/adoption-and-usage>
- Some people haven't gotten the message, or slip up, allowing local networks to mess about; <https://whynohhttps.com/> has lists, including by country
- One way to mess up: replace an old service that used http:// URLs with one that uses https:// URLs but leave references to the old thing in place and the old service running (even with a mandatory re-direct to the new one)
 - <https://www.irishimmigration.ie/wp-content/uploads/2021/04/INIS-online-portal-guidance.pdf> contains a link to the "old" thing here which was <http://www.inis.gov.ie> (accessed 2026-09-16)
- Browser warnings (interstitial pages) may save you, but not if you click-through those
 - If you used the old service URL a lot, you might get used to doing that click-through

Fake a real web site

- A legacy Irish government web site: inis.gov.ie
- Inside my laptop I'm running a web server
- I can “easily” re-direct the traffic for the insecure URL to that
 - This might not work for you, depends if their IP addresses have changed and what DNS you use, and if I mucked up, but we'll see...

```
# start lighttpd
$ ./testlighttpd.sh
$ sudo sysctl -w net.ipv4.conf.all.route_localnet=1
$ sudo iptables -t nat -A PREROUTING -i
enx38c986f09157 \
    -p tcp -d 146.0.55.149 --dport 80 \
    -j DNAT --to-destination 127.0.0.1:8099
```